VIRINCHI SAI ATHMAKURI

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Professional Summary

Applied AI & Machine Learning Engineer with 2+ years of experience architecting secure AI systems, enterprise LLM platforms, Retrieval-Augmented Generation (RAG) applications, Model Context Protocol (MCP) integrations, and cloud-native AI infrastructure. Experienced designing production-oriented AI software spanning intelligent document systems, enterprise knowledge platforms, AI security, adversarial machine learning, deepfake detection, and agentic AI workflows using Python, FastAPI, PyTorch, LangChain, PostgreSQL, Docker, Kubernetes, AWS, and GCP.

My work focuses on translating applied AI research into production-ready software by engineering secure AI platforms, scalable retrieval systems, intelligent automation, and enterprise AI infrastructure with strong emphasis on system architecture, observability, reliability, deployment, and responsible AI.

Technical Expertise

Applied AI	Enterprise LLMs, Retrieval-Augmented Generation (RAG), LangChain, LlamaIndex, Prompt Engineering, Context Engineering, Hugging Face, AWS Bedrock, Vertex AI, OpenWebUI
LLM Platform Engineering	Model Context Protocol (MCP), Agentic AI, Tool Calling, Multi-Agent Systems, AI Orchestration, Knowledge Retrieval, Semantic Search, AI Platform Architecture
Machine Learning	PyTorch, TensorFlow, Scikit-learn, CNN, CNN-LSTM, Autoencoders, Transfer Learning, Embeddings, Feature Engineering, Model Evaluation
AI Security	Adversarial Machine Learning, FGSM, PGD, Deepfake Detection, Prompt Security, Secure AI Systems, Model Robustness, AI Safety
Backend Engineering	Python, FastAPI, Flask, REST APIs, PostgreSQL, PGVector, SQL, Bash, C/C++, API Design, Authentication
Cloud Platform Engineering	AWS, GCP, Docker, Kubernetes, Terraform, Linux, Git, GitHub Actions, Jenkins, Cloud Deployment
Data Engineering	PostgreSQL, MySQL, PGVector, FAISS, Pinecone, Weaviate, Qdrant, Kafka, Airflow, ETL Pipelines
MLOps	MLflow, Kubeflow, ONNX Runtime, CI/CD, Model Serving, GPU Inference, Experiment Tracking, Deployment Automation
Observability	Prometheus, Grafana, Splunk, ELK, CloudWatch, ServiceNow, Monitoring, Logging, Alerting

Professional Experience

State University of New York Polytechnic Institute — Utica, NY

GenAI Research Engineer — Jan 2026 – Present

- Designed and developed an Intelligent Research Assistant that enables semantic document retrieval, contextual question answering, technical summarization, and grounded reasoning across research publications, institutional documents, and engineering knowledge bases.
- Architected Retrieval-Augmented Generation (RAG) pipelines using LangChain, LlamaIndex, FastAPI, PostgreSQL, and PGVector to support secure document ingestion, embedding generation, semantic indexing, hybrid retrieval, and grounded response generation.
- Engineered modular AI services for document parsing, chunking, metadata extraction, vector indexing, prompt orchestration, and LLM inference, enabling reusable enterprise AI workflows for research environments.
- Developed transformer-based machine learning pipelines using PyTorch and TensorFlow for experimentation, secure inference, model evaluation, and reproducible AI research across multiple cybersecurity-focused applications.
- Built scalable Python-based research workflows for preprocessing, experimentation, evaluation, visualization, and technical documentation while improving reproducibility and engineering efficiency across collaborative AI projects.
- Collaborated on applied AI research involving secure AI systems, adversarial robustness, Retrieval-Augmented Generation, and trustworthy AI for cybersecurity-oriented applications under the supervision of Dr. Mahmoud Badr.

GenAI Research Engineer — Jul 2025 – Dec 2025

- Architected and implemented a secure on-premises enterprise LLM platform using OpenWebUI, FastAPI, LangChain, and Retrieval-Augmented Generation (RAG) to enable GPT-style interactions over institutional knowledge without reliance on external cloud-hosted models.
- Designed secure retrieval workflows using embedding models, vector databases, RBAC-based authorization, semantic search, and grounded prompting to improve response accuracy while minimizing hallucinations and protecting sensitive institutional information.
- Implemented modular backend services supporting document ingestion, embedding generation, retrieval orchestration, prompt construction, response generation, and enterprise knowledge management using Python-based microservice architecture.
- Evaluated prompt injection attacks, jailbreak attempts, adversarial prompts, and LLM security vulnerabilities to improve model resilience, prompt robustness, and secure deployment practices for private AI systems.
- Collaborated on API authentication, encryption-aware system design, network hardening, secure deployment planning, and containerized infrastructure using Docker to support privacy-preserving AI environments.

- Improved inference efficiency by evaluating model optimization strategies including vLLM-based serving, quantization concepts, and GPU-enabled deployment approaches for scalable enterprise AI workloads.

AI Research Engineer Feb 2025 – Jul 2025

- Conducted interdisciplinary AI research supporting intelligent sensing systems, biomedical applications, and IoT-enabled self-powered devices through investigation of oxide-based triboelectric nanogenerator (TENG) technologies.
- Applied AI-assisted research workflows using Python for literature mining, technical analysis, preprocessing, comparative evaluation, and engineering documentation across multidisciplinary research initiatives.
- Performed comparative analysis of ZnO and TiO_2 based triboelectric materials to evaluate energy harvesting efficiency, charge generation mechanisms, material stability, and biomedical sensing potential.
- Strengthened interdisciplinary engineering workflows by integrating intelligent data analysis, technical research automation, reproducible experimentation, and scalable documentation processes supporting applied research.

AI/ML Research Engineer Aug 2023 – Jul 2024

- Developed deepfake audio detection models using PyTorch, transfer learning, and deep neural networks to distinguish authentic and synthetic speech across forensic AI datasets.
- Improved dataset preprocessing, feature extraction, model regularization, hyperparameter experimentation, and evaluation workflows to enhance robustness, interpretability, and generalization of deep learning models.
- Investigated adversarial considerations, dataset limitations, and emerging attack techniques affecting synthetic speech detection while evaluating practical challenges in forensic AI systems.
- Co-authored a comprehensive research survey reviewing more than thirty deepfake audio detection approaches, benchmark datasets, evaluation methodologies, and open research challenges in AI-driven digital forensics.

ConnX AI Inc. Hyderabad, India

Data & Cloud Engineer Apr 2023 – Jul 2023

- Engineered cloud observability solutions using Python, Prometheus, Nagios, and Zabbix to monitor large-scale UCaaS and Session Border Controller (SBC) environments, improving operational visibility and reducing incident frequency by approximately 50%.
- Developed a real-time log analytics and anomaly detection platform using Flask, enabling continuous telemetry ingestion, intelligent alert generation, and operational monitoring across geographically distributed infrastructure.
- Automated IP reputation analysis, blacklist recovery, routing diagnostics, and infrastructure health checks through Python-based REST automation, reducing manual operational effort and accelerating issue resolution.
- Containerized monitoring applications with Docker and designed CI/CD pipelines using Jenkins and GitHub Actions to improve deployment consistency, release reliability, and infrastructure reproducibility.
- Built centralized operational dashboards integrating AWS CloudWatch, Prometheus, and SBC telemetry to support SLA reporting, KPI visualization, latency analysis, and production incident monitoring.

Selected AI Systems

SentinelAI — Enterprise AI Workspace

MCP-Powered Secure AI Platform for Enterprise Knowledge & Tool Orchestration

- Architected and developed a self-hostable enterprise AI workspace implementing the Model Context Protocol (MCP) to securely connect large language models with enterprise tools, internal knowledge repositories, developer workflows, and structured business data.
- Engineered a modular platform using Next.js, FastAPI, PostgreSQL/PGVector, and Retrieval-Augmented Generation (RAG) to enable semantic search, contextual reasoning, grounded AI responses, and enterprise knowledge discovery across private datasets.
- Built multiple MCP servers exposing GitHub, Gmail, Google Calendar, Filesystem, PostgreSQL, and document retrieval capabilities through standardized interfaces, enabling secure AI-assisted task execution and intelligent workflow automation.
- Implemented enterprise-grade authentication and governance using JWT, Role-Based Access Control (RBAC), audit logging, sandboxed filesystem access, SQL safety enforcement, prompt security protections, and approval gates for high-risk operations.
- Designed production-ready deployment architecture using Docker, Kubernetes, Prometheus, GitHub Actions, CodeQL, automated testing, and security regression workflows emphasizing observability, maintainability, and enterprise scalability.
- Applied modular software architecture principles separating authentication, orchestration, retrieval, AI inference, MCP integration, monitoring, and persistence services to improve extensibility, testing, and long-term maintainability.

SecureAI SOC Copilot

Enterprise GenAI Assistant for Security Operations

- Designed and developed a secure GenAI copilot enabling Security Operations Center (SOC) analysts to perform evidence-based investigations over enterprise security documentation using Retrieval-Augmented Generation (RAG).
- Engineered modular FastAPI services supporting document ingestion, semantic chunking, embedding generation, FAISS indexing, retrieval orchestration, prompt construction, and grounded response generation.
- Implemented secure authentication, role-aware document isolation, SQLite audit logging, OpenAI-compatible model integration, and enterprise document management workflows supporting secure AI adoption.
- Architected the platform using independent authentication, retrieval, inference, audit, and document management layers to improve scalability, maintainability, and future extensibility.
- Enabled contextual security investigations, intelligent document search, and evidence-backed AI responses while preserving enterprise security boundaries and organizational knowledge governance.

Secure Agentic CloudOps SIEM Platform
AI Platform for Cloud Security Operations

- Designed an event-driven cloud security platform supporting centralized log ingestion, intelligent detection, alert persistence, and AI-assisted investigation across cloud-native infrastructure.
- Engineered modular FastAPI microservices using PostgreSQL, Docker, and Kafka-compatible event streaming to support scalable event processing, alert management, and future autonomous remediation workflows.
- Architected loosely coupled ingestion, detection, persistence, investigation, and orchestration services to improve scalability, maintainability, and operational resilience for enterprise cloud environments.
- Designed the platform for integration with AI-assisted incident investigation, alert triage, automated runbooks, and future agent-driven remediation across hybrid cloud deployments.
- Applied production-oriented software engineering practices including containerization, API-first design, modular architecture, observability planning, and secure infrastructure design.

PageMint Offline
Privacy-First Local AI Document Intelligence Platform

- Developed a privacy-first offline document intelligence platform enabling secure document conversion, local AI preprocessing, and structured knowledge extraction without transmitting sensitive files to external AI services.
- Implemented localhost-only execution, temporary file cleanup, configurable request limits, secure file handling, and privacy-preserving processing to support enterprise document security requirements.
- Built document transformation pipelines producing structured Markdown output optimized for downstream Retrieval-Augmented Generation (RAG), semantic indexing, and enterprise knowledge management workflows.
- Designed a lightweight modular architecture supporting future AI-powered document understanding, enterprise search, and secure local knowledge processing.

Research Contributions

GACL-Fusion: Hybrid ML/DL Framework for Adversarially Robust Intrusion Detection
Master's Thesis 2025
Advisor: Dr. Zahid Akhtar

- Designed a hybrid intrusion detection framework integrating genetic algorithm-based clustering, CNN-LSTM architectures, denoising autoencoders, and ensemble learning techniques to improve resilience against adversarial cyber attacks.
- Evaluated model robustness against FGSM and PGD adversarial attacks using NSL-KDD and UNSW-NB15 benchmark datasets while improving rare and zero-day attack detection through automated feature engineering and adaptive model training.
- Investigated practical deployment considerations including scalability for edge environments, online learning strategies, explainable AI techniques, and production-oriented cybersecurity applications.

Selected Research Publications

- Video and Audio Deepfake Datasets and Open Issues in Deepfake Technology — Forensic Sciences (2024)
- Design and Implementation of an AI Virtual Mouse Using Hand Gesture Recognition
- Scarlett: Virtual Assistant & Browlett Browser
- An Intelligent Way to Recognize Digits using Convolutional Neural Networks
- Tachyon: Bike Rentals Made Easy

Education

Master of Science — Network and Computer Security
State University of New York Polytechnic Institute
Bachelor of Engineering — Computer Science
Methodist Engineering College

Certifications

- CompTIA Security+ (SY0-701)
- AWS Academy Cloud Foundations
- Google Cloud Ready Facilitator (6 Skill Badges)
- Cisco DevNet
- Cisco Cybersecurity Essentials
- Cisco Networking Essentials

Open Source

github.com/virinchisai — Building secure enterprise AI systems, Model Context Protocol (MCP) platforms, Retrieval-Augmented Generation (RAG) applications, AI security solutions, cloud-native AI infrastructure, and privacy-first local AI utilities.
Featured Repositories: SentinelAI — Enterprise AI Workspace; SecureAI SOC Copilot; Secure Agentic CloudOps SIEM Platform; PageMint Offline.